INVESTIGATING MAX PRESSURE JUMP IN MIXED PHASE

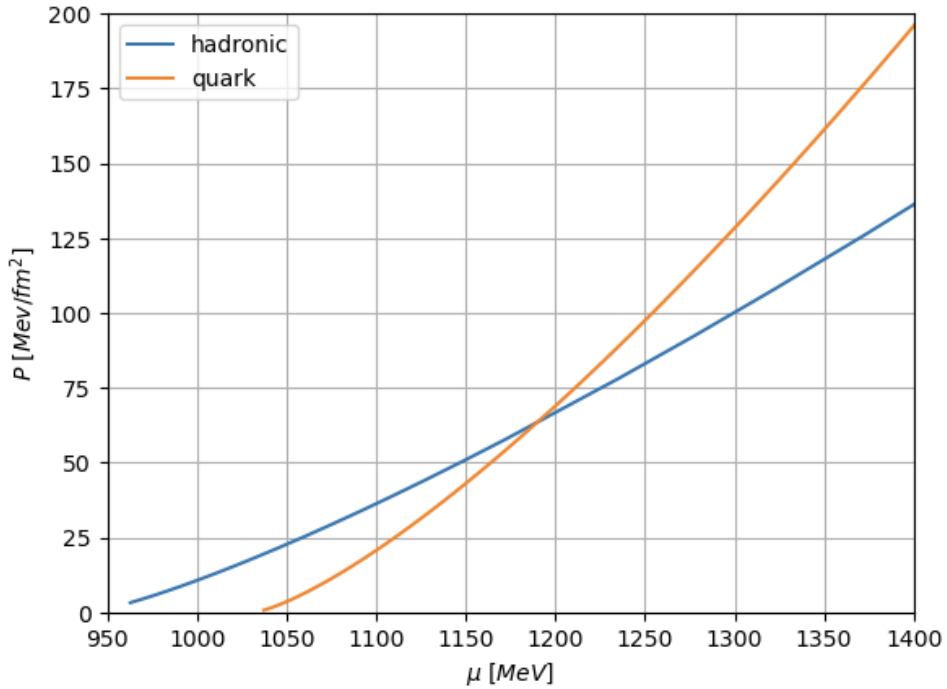
Katta Naveen

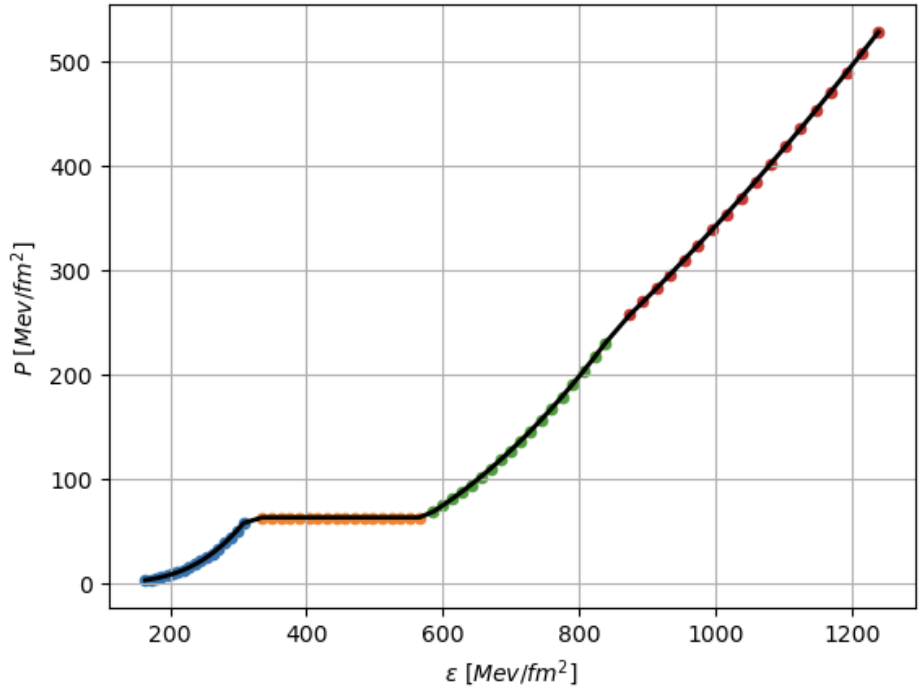
Supervisor : Dr .Ritam Mallick



INTRODUCTION

- We investigate the phase transition between hadronic to quark matter in the core of neutron stars by using a quadratic interpolation method.
- By implementing a piecewise polytropic equation of state (EOS) for the hadronic and quark phases, we create a smooth transition across the mixed phase.
- We are trying to learn how the smoothness of the transition behaves if we keep on changing the EoS.
- And finally for a any given EoS can we determine the maximum smoothness of the mixed phase.



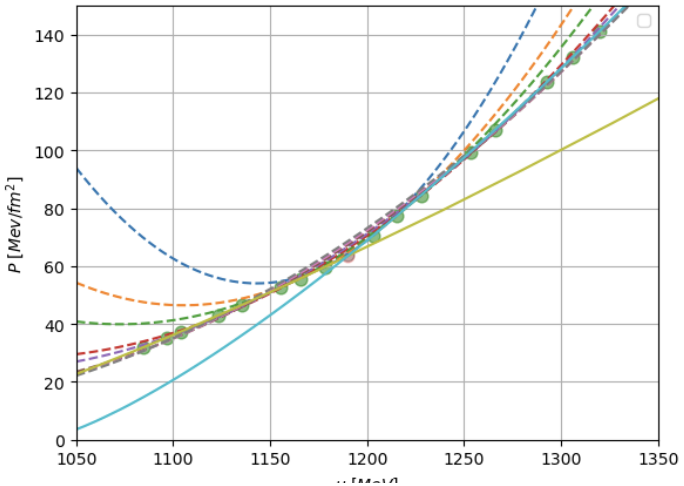


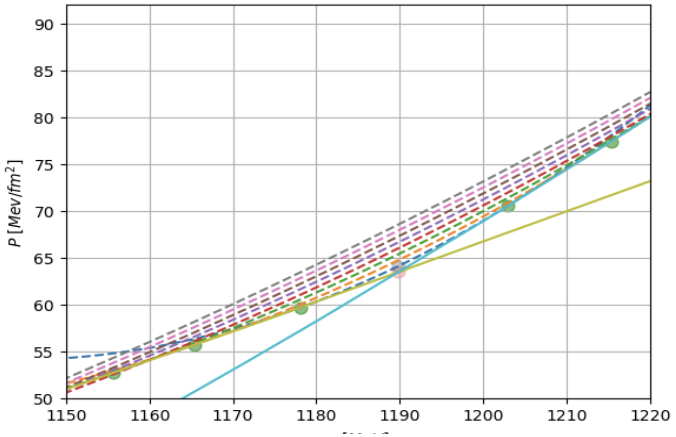
CONSTRUCTING EoS

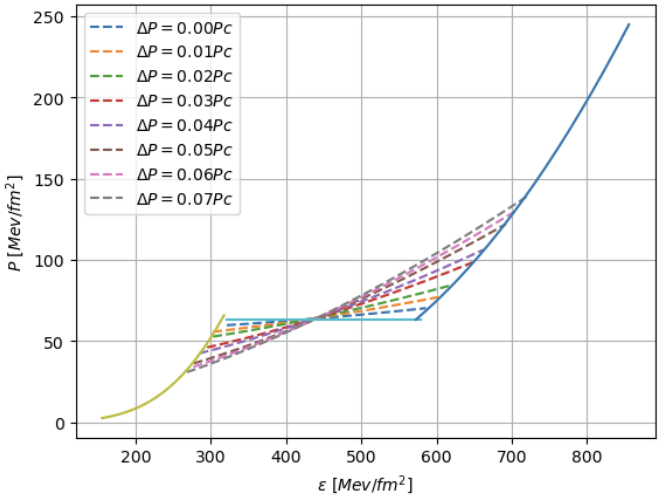
- We construct EoS using Piecewise Polytropes.
- EoS is divided into segments, each with its own polytropic index (Gamma) and number densities, describing distinct phases (hadronic, quark).

	Γ_3	κ_3 [MeV fm$^{3(\Gamma_3-1)}$]	m_0 [MeV]
i = 1	4.921	17906.60	939.56
i = 2	0	63.177	1201.75
i = 3a	4.00	774.375	1031.815
i = 3b	2.80	548.309	958.553

- $P(n) = n^2 \frac{d\left(\frac{\varepsilon(n)}{n}\right)}{dn}; \frac{\varepsilon(n)}{n} = \int dn \frac{P(n)}{n^2} = \int dn\ \kappa n^{\Gamma-2} = \frac{\kappa n^{\Gamma-1}}{\Gamma-1} + C,$
- $\mu(n) = \frac{P(n)+\varepsilon(n)}{n} = \frac{\kappa\Gamma}{\Gamma-1} n^{\Gamma-1} + m_0.$







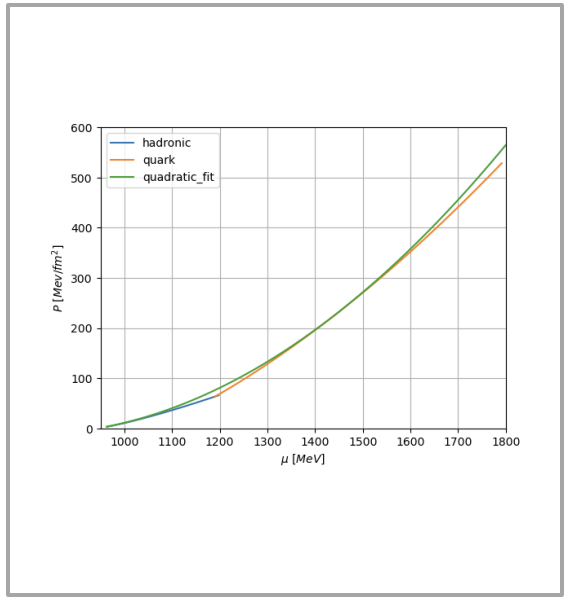
QUADRATIC INTERPOLATION

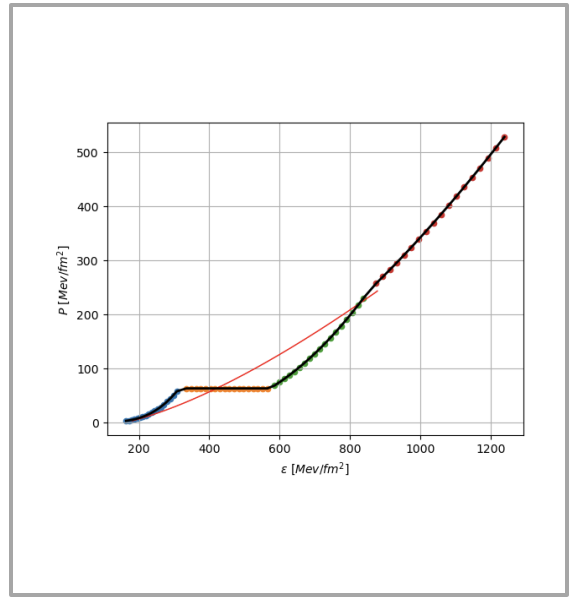
- We consider quadratic form of the mixed phase around a pressure point, $P_M(\mu) = \alpha(\mu - \mu_c)^2 + \beta(\mu - \mu_c)^1 + P_c + \Delta P$.
- We can find the alpha and beta from the below equations
- $$\alpha = \frac{1}{\mu_{cQ} - \mu_{cH}} \left(\frac{P_Q(\mu_{cQ}) - (\Delta P + P_c)}{\mu_{cQ} - \mu_c} - \frac{P_H(\mu_{cH}) - (\Delta P + P_c)}{\mu_{cH} - \mu_c} \right)$$
- $$\beta = \frac{P_H(\mu_{cH}) - (\Delta P + P_c)}{\mu_{cH} - \mu_c} + \frac{P_Q(\mu_{cQ}) - (\Delta P + P_c)}{\mu_{cQ} - \mu_c} - \frac{P_Q(\mu_{cQ}) - P_H(\mu_{cH})}{\mu_{cQ} - \mu_{cH}}$$
  - μ_{cH} and μ_{cQ},

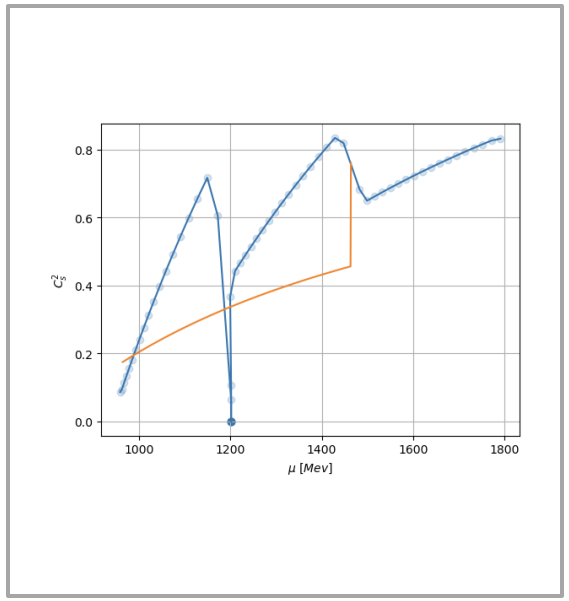
- Are the respective chemical potential where transition occurs in the mixed phase.

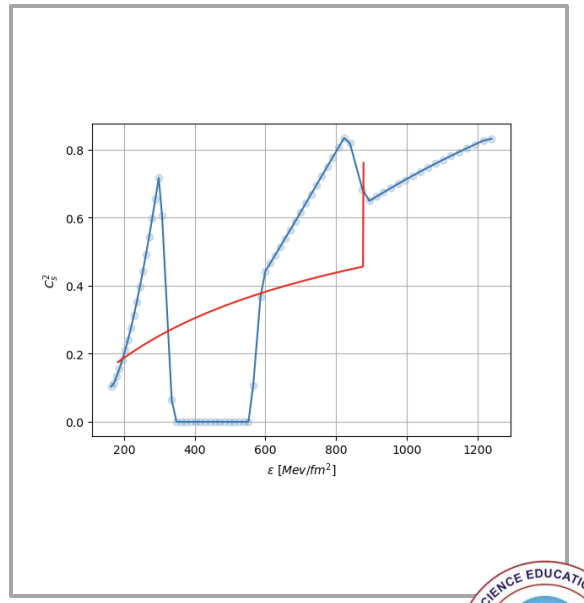
MAXIMUM INTERPOLATION ACHIEVABLE

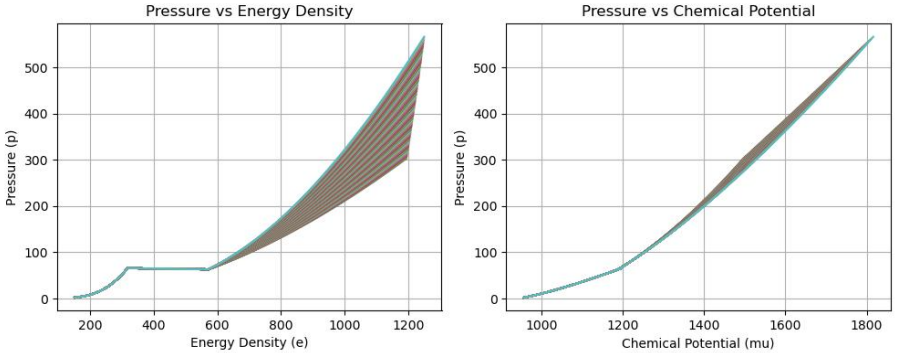
- Now we will keep the initial transition chemical potential to initial chemical potential of hadronic phase.
- We develop a model such that it finds the optimal values of alpha and beta for such EoS.
- Constraints of the Model : Equality constraint and Inequality constraint.

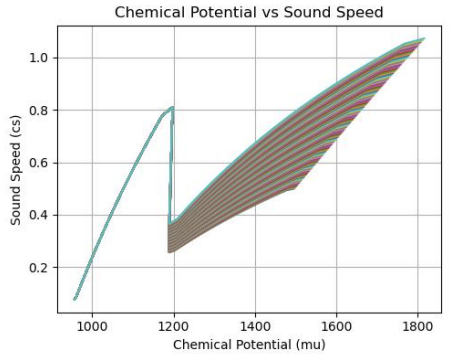


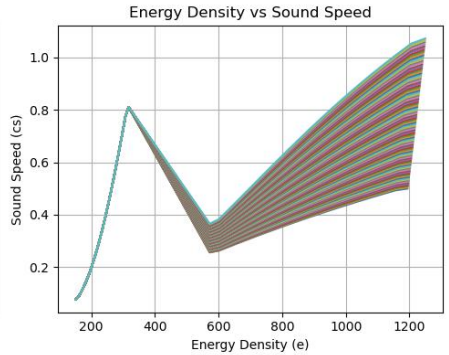


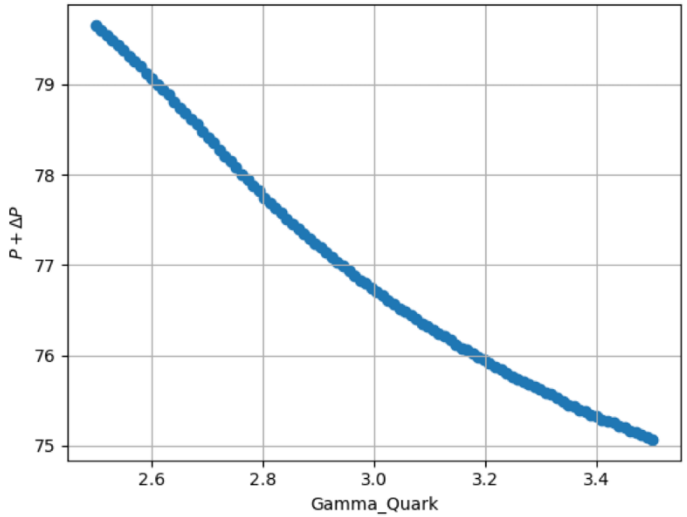










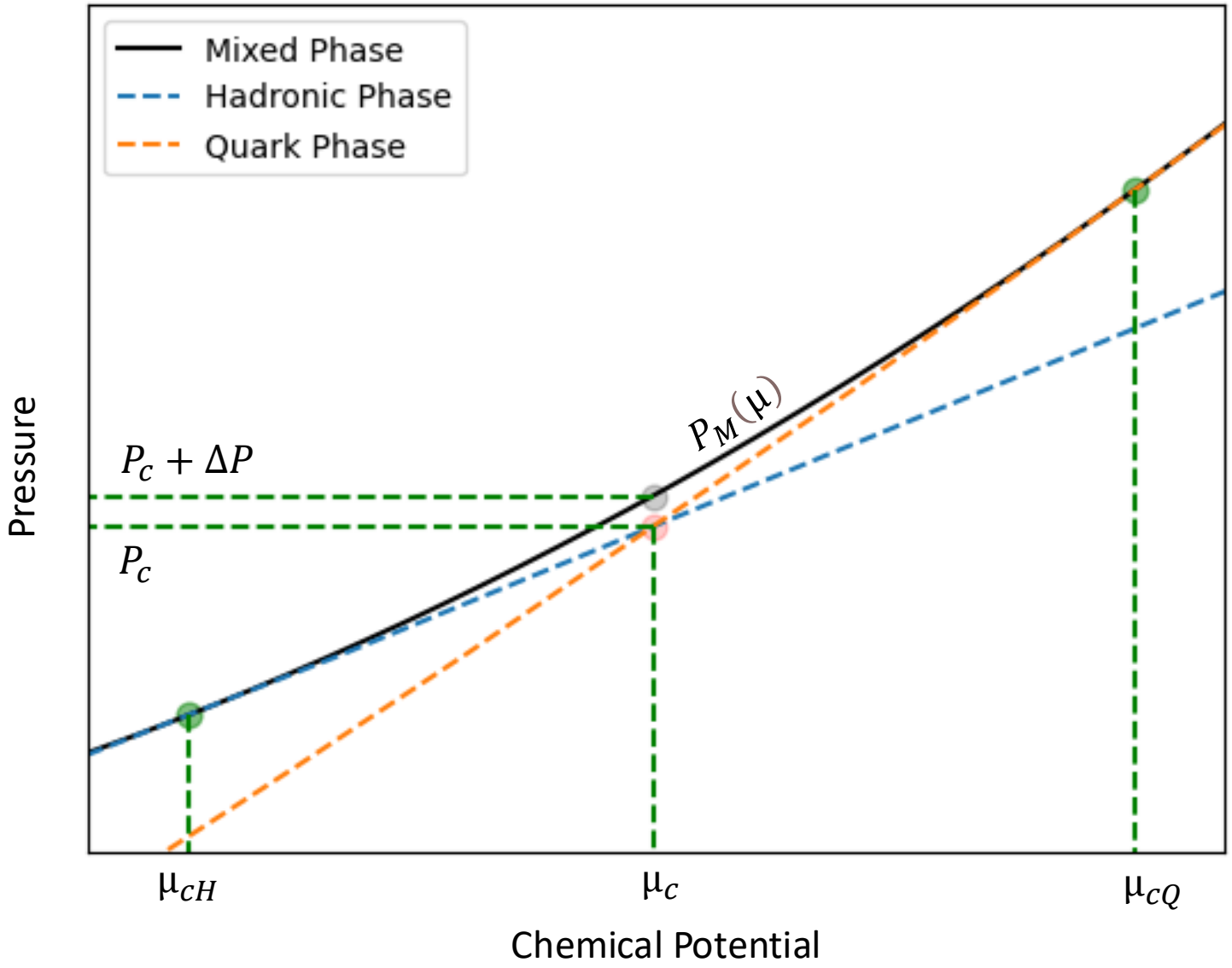


IMPACT OF DIFFERENT POLYTROPIC INDEX

- If we change the Gamma value of Quark, Polytrope changes so does the EoS, Gamma Quark is ranging from 2.5 to 3.5 .
- Now if we find the maximum interpolation for these different EoS, we get the respective additional pressure terms.
- So in the figure we can see that if we keep on increasing the polytrope index of quark the additional pressure term is keep on decreasing following a trendline.



THANK YOU


Mixed Phase
Hadronic Phase
Quark Phase
$P_M(\mu)$
$P_c + \Delta P$
P_c
Pressure
μ_{cH}
μ_c
μ_{cQ}
Chemical Potential